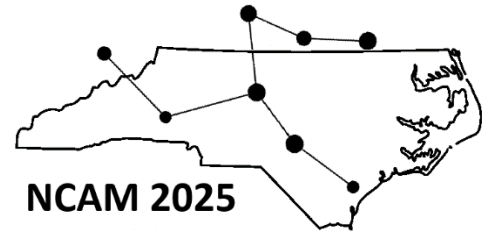


NC Astronomers' Meeting

Saturday, 27 September 2025

Koury Auditorium, Guilford Technical Community College
621 East Main St., Jamestown, NC, 27282



8:45 a.m. Conference Opens

Refreshments are available throughout the day in the display area.

9:20 a.m. Welcome & Announcements

9:30 a.m. Invited Speaker

Grant Tremblay (Center for Astrophysics | Harvard & Smithsonian) ***Galaxies across the Gulf of Time***

Galaxies are impossibly vast kingdoms of dark matter, stars, gas, dust, and planets beyond imagining. I'll give a tour of some of the most jaw-dropping galaxies in the known universe, and talk about our understanding of galaxies in the context of our human story of discovery, from the first stargazers 30,000 years ago who carved the heavens into art, to the revolution of telescopes that revealed the Milky Way as but one among countless galaxies, and now to the transformative breakthroughs of the James Webb Space Telescope. This talk will trace how humanity's expanding vision has reshaped our place in the cosmos, illuminating not only what galaxies are, but also what they reveal about our shared journey to understand the universe.

Dr. Grant Tremblay is an Astrophysicist and Director of External Relations at the Center for Astrophysics | Harvard & Smithsonian, and Vice President of the American Astronomical Society (AAS). Previously, he was a NASA Einstein Fellow at Yale University, a Fellow at the European Southern Observatory (ESO) near Munich, and an Astronomer at ESO's Very Large Telescope in Chile. He has authored more than 100 publications in astronomical journals and three books for the general public, including *The Sky and Other Ghosts*, coming next year from Princeton Press. He frequently appears on Science and Discovery Channel documentary series including *How the Universe Works*, as well as in recent and forthcoming projects with the BBC and PBS NOVA.



10:30 a.m. Break

10:45 a.m. Contributed Oral Session I

- 1.1 Alp Tezbasaran, NC Museum of Natural Sciences
Spiral Graph: Cluster Buster - Pairing Citizen Scientists with A.I. to Identify Individual Arms in Spiral Galaxies
- 1.2 Anatoly Miroshnichenko, UNC-Greensboro
LN Hya - unexpected phenomena in a seemingly well-studied star
- 1.3 Noga Luna Wasserberg, NC State University
Revealing The Nature of The Misclassified Star HD 59509
- 1.4 Delilah Gates, Center for Astrophysics, Harvard & Smithsonian
Morphology of Relativistically Broadened Emission Lines around Accreting Black Holes
- 1.5 Edoardo Levati, Wake Forest University
Cumulative effect of orbital resonances in extreme-mass-ratio inspirals
- 1.6 Alejandro Cárdenas-Avendaño, Wake Forest University
Photon Rings as Probes of Strong Gravity: Toward Space-Based Black Hole Imaging with BHEX

12:15 p.m. Lunch – on your own – there are many options near the GTCC campus.

- 2:00 p.m. *Contributed Oral Session II***
- 2.1 Josh Haislip, UNC-Chapel Hill
Special Session – Skynet 2
 - 2.2 Dan Caton, Appalachian State University
DarkSky International - NC Chapter: Recent Activities
 - 2.3 Ian Hewitt, Coastal Carolina University
Universities, Cubesats, and Astronomy
 - 2.4 Enrique Gómez, Western Carolina University
What is the Feynman Question for teaching Astro101?

3:15 p.m. *Final Announcements/Break*

3:30 p.m. *Astro 101 Teaching Exchange – Open Discussion*

No Display presentations this year

Presentation Abstracts

10:45 a.m. *Contributed Oral Session I*

1.1 Alp Tezbasaran, NC Museum of Natural Sciences

Spiral Graph: Cluster Buster - Pairing Citizen Scientists with A.I. to Identify Individual Arms in Spiral Galaxies

We introduce Spiral Graph: Cluster Buster, an online citizen science project built on the Zooniverse platform that is dedicated to identifying individual arms of spiral galaxies from the precursor Spiral Graph project. In Spiral Graph, volunteers were asked to trace the observed spiral structure of thousands of galaxies in order to artificially increase the signal-to-noise ratio of the low contrast images. Spiral Graph: Cluster Buster seeks to untangle the aggregate tracings provided by multiple volunteers for each galaxy by training an A.I. algorithm, with the help of citizen scientists, to cluster tracings that represent individual arms. The goal of this project is to quantitatively measure the winding of individual arms, a parameter known as pitch angle, in order to take advantage of the pitch angle-central black hole mass relationship and identify any potential intermediate mass black hole candidates in nearby spiral galaxies.

1.2 Anatoly Miroshnichenko, UNC-Greensboro

***S. Achkassova, S. Nurmakhametova, S. Danford, V.G. Klochkova, S.V. Zharikov, N.L. Vaidman, S.A. Khokhlov
LN Hya - unexpected phenomena in a seemingly well-studied star***

LN Hya is a 6-th magnitude spectral type F supergiant in the southern constellation Hydra located far from the Galactic plane. Initial studies of its spectrum suggested a deficit of many chemical elements in the star's atmosphere that is consistent with a low-mass star at an advanced evolutionary stage. The star's surface temperature was found to be close to that of the Sun. However, the first spectrum taken at TCO 4 years ago revealed the presence of molecular bands, which appear at a much lower surface temperature. Follow-up observations showed that the molecular bands disappeared, but atomic spectral lines were found to change their positions and strengths with a stable period of 148.5 days. Brightness variations followed the same period for over last 10 years but were not detected 40 years ago. These phenomena will be briefly described along with possible interpretations.

1.3 Noga Luna Wasserberg, NC State University

***Anatoly Miroshnichenko, Steve Danford
Revealing The Nature of The Misclassified Star HD 59509***

HD 59509 is an 8-magnitude star located in the middle of the constellation Gemini. It was originally classified as a spectral type F8 star which is just a little hotter than our own Sun. However, already first spectra taken at high resolution showed hydrogen emission lines and other spectral features of a much hotter star. Monitoring of the star's spectrum in the last quarter century indicated a gradually increasing surface temperature, which photometric surveys showed an optical brightness decrease. This is a typical behavior of a dying star of a star with a mass of a few times that of the Sun, but in this case, it turned out to be more complicated. We review the main results of TCO spectroscopy and brightness variations and comment on a possible nature of the star.

1.4 Delilah Gates, Center for Astrophysics, Harvard & Smithsonian

Amrita Sahu, Chau Truong, & Alejandro Cárdenas-Avendaño

Morphology of Relativistically Broadened Emission Lines around Accreting Black Holes

Single-frequency emission from an accretion disk around a black hole is broadened into a line profile due to gravity and the motion of the disk relative to the observer. In this talk, I present unified analytical explanation for the line profile morphological features (i.e., its extent, kinks, and fall-off), and show how these features can be used to constrain black hole spin, observer inclination, and size of the disk.

1.5 Alejandro Cárdenas-Avendaño, Wake Forest University

Photon Rings as Probes of Strong Gravity: Toward Space-Based Black Hole Imaging with BHEX

One of the most striking lensing features predicted by general relativity is the black hole photon ring—a thin, bright ring of light produced by photons that orbit multiple times around the black hole. These rings encode direct information about a black hole's mass, spin, and surrounding environment. In this talk, I will introduce the concept of black hole photon rings, explain why they provide a unique window into fundamental physics, and describe how the proposed Black Hole Explorer (BHEX) mission aims to measure them with unprecedented precision. By extending the groundbreaking work of the Event Horizon Telescope into space, BHEX will enable sharp, high-resolution imaging of M87* and Sgr A*, opening new opportunities to probe the nature of strong gravity.

1.6 Edoardo Levati, Wake Forest University

Alejandro Cárdenas-Avendaño

Cumulative effect of orbital resonances in extreme-mass-ratio inspirals

Orbital resonances in extreme-mass-ratio inspirals (EMRIs) have been proven to be a key feature for accurate gravitational-wave template modeling. Decades of research have led to schemes that can not only model the adiabatic inspiral of such a binary system, but also account for the effects of resonances on their evolution. In this work, we use an effective resonance model that includes analytically derived corrections to the radiation reaction fluxes, to study the combined effects of both dominant (low-order) and sub-dominant (high-order) orbital resonances in EMRIs. We show that using single, universal shifts for all fluxes overestimates the resonance impact, and therefore individualized shifts for each resonance crossing are needed for accurate modeling. Our analysis reveals that the cumulative effects from multiple resonance crossings can significantly impact the orbital evolution of EMRIs, especially for highly eccentric orbits. Our results provide further evidence that resonance effects have to be included in template production to extract detailed astrophysical parameters from EMRI signals.

10:45 a.m. Contributed Oral Session II

2.1 Josh Haislip, UNC- Chapel Hill

SkyNet 2 Demonstration

2.2 Dan Caton, Appalachian State University

DarkSky International - NC Chapter: Recent Activities

We report on our chapter's efforts across the state to reduce light pollution. These included projects and pending projects in Ashe County, along the Blue Ridge Parkway, Hyde County, and in Raleigh. Activities at the American Astronomical Society's COMPASSE committee and its Campus SHINE program will be reported. Anyone at NCAM who has light pollution activities will be invited to share them.

2.3 Ian Hewitt, Coastal Carolina University

Universities, Cubesats, and Astronomy

Many universities are working on or in projects to launch small satellites to build space experience and do science. Initially most of the satellites are pointed toward the Earth, but there is a new generation of projects that will use cubesats to do astronomy.

2.4 Enrique Gómez, Western Carolina University

What is the Feynman Question for teaching Astro101?

Higher Education is undergoing a major structural change driven by demographics, budget rescission, and a societal reevaluation of the college degree. College introductory astronomy courses (Astro 101), as part of a general education requirement for most institutions, will likely have its standing challenged as academic programs are downsized or eliminated. Given these along with cuts in NASA and NSF support for research and student experiences, how should we prioritize the content of the Astro 101 course as well as frame its value to students and society? We revisit Richard Feynman's thought experiment about what would be a fundamental statement that would allow all of modern science to be reconstructed. In the same vein we pose the question: what are the fundamental statements, that if passed on, would allow all of modern astronomy to be rebuilt? We present multiple, non-exclusionary, approaches to this question and invite a community discussion.